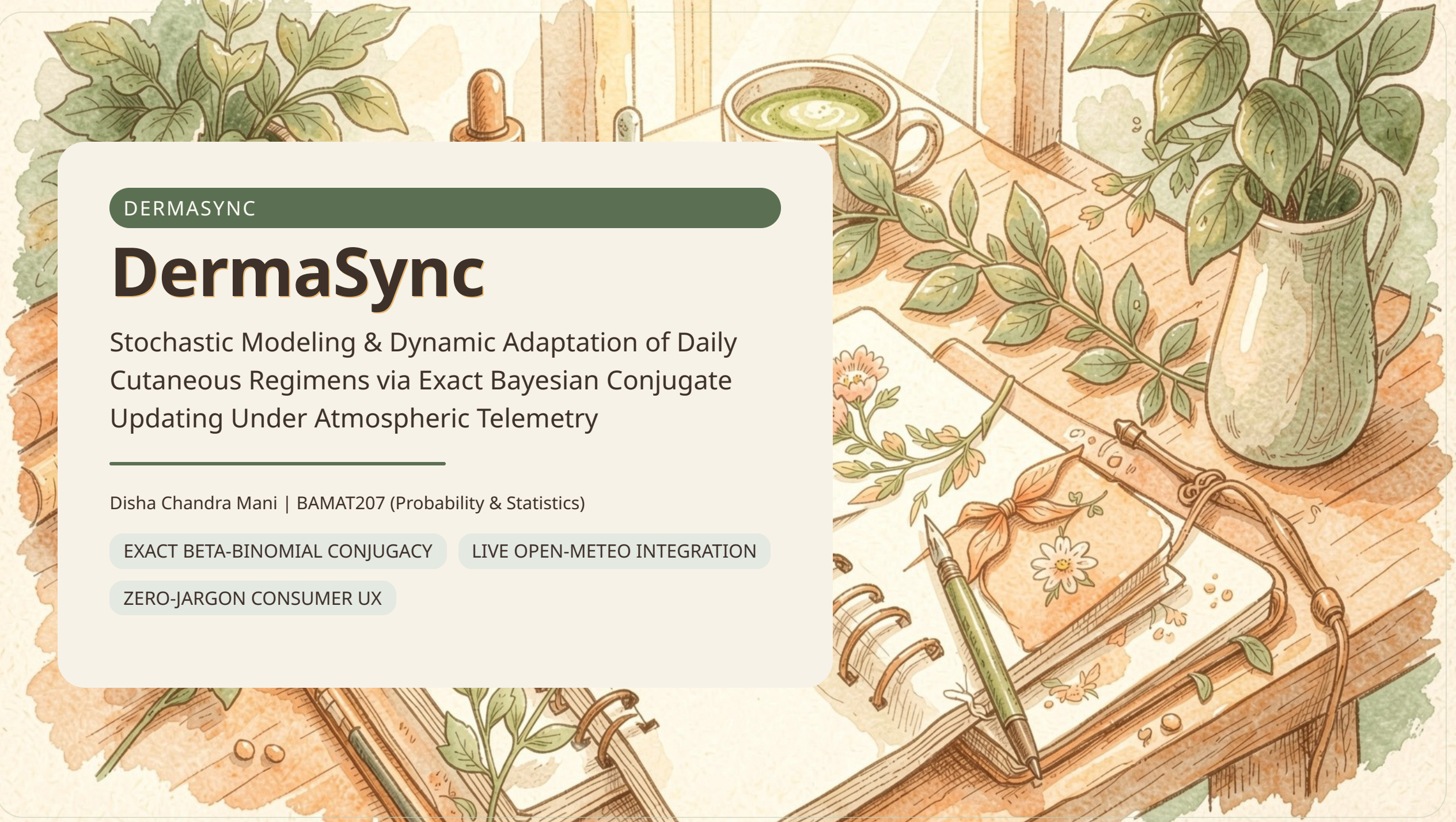


DERMASYNC

DermaSync

Stochastic Modeling & Dynamic Adaptation of Daily Cutaneous Regimens via Exact Bayesian Conjugate Updating Under Atmospheric Telemetry

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EXACT BETA-BINOMIAL CONJUGACY

LIVE OPEN-METEO INTEGRATION

ZERO-JARGON CONSUMER UX



Extreme humidity spikes (monsoon/summer >80% RH) stimulate excess sebum and follicular congestion under heavy occlusives.



Dry heat / arid drops (<40% RH) accelerate Transepidermal Water Loss (TEWL), rendering exfoliating acids acutely irritating.

Traditional skincare routines ignore the day’s microclimate

- The Everyday Frustration: Traditional skincare routines are rigidly static (“apply 2% BHA and cream daily”), ignoring microclimate shifts and daily skin variation.

- Beginners do not know when to buffer, swap, or dial back actives, leading to barrier damage and routine abandonment.

THE DAILY LOOP

DermaSync turns weather and one-tap feedback into a daily routine

An intuitive, low-friction daily routine companion personalizes AM/PM skincare steps using live weather telemetry and 1-tap user feedback, updating recommendations over time via Bayesian learning.

01

Automated Environmental Ingestion

Read live weather (RH, Temp, UV) via Open-Meteo REST API.

02

5-Second Check-In

Replace 20-question derm forms with a 1-tap feel selector (Balanced, Tight/Dry, Greasy/Clogged, Stinging).

03

Dynamic Routine Synthesis

Output concrete, weather-adjusted steps (Cleanse, Treat, Moisturize, Protect).

04

Closed-Form Bayesian Learning

Update the user’s latent tolerance parameter θ over time.

MATHEMATICAL FRAMEWORK

Exact Beta-Binomial updating makes tolerance learning transparent

- Latent Parameter: $\theta \in [0, 1]$ represents the user's true tolerance/satisfaction rate for a routine archetype.
- Prior: $\theta \sim \text{Beta}(\alpha_0, \beta_0)$, where $E[\theta] = \alpha_0 / (\alpha_0 + \beta_0)$.
- Likelihood: over n trial days, k "Balanced" check-ins follow $k \mid \theta \sim \text{Binomial}(n, \theta)$.
- Posterior: $\theta \mid k \sim \text{Beta}(\alpha_1, \beta_1) = \text{Beta}(\alpha_0 + k, \beta_0 + n - k)$.
- Posterior variance: $\text{Var}(\theta) = (\alpha_1\beta_1) / [(\alpha_1 + \beta_1)^2(\alpha_1 + \beta_1 + 1)]$.

Completely closed-form and exact; bypasses stochastic MCMC error and approximation latency.

NUMERICAL STABILITY

Log-space calculus and dew point computation keep the engine stable

Lanczos 7-Term Log-Gamma Approximation

evaluates ln Γ(z) and log-Beta ln B(a, b) in log-space to prevent floating-point underflow during continuous density calculations.

$$\ln B(a,b) = \ln \Gamma(a) + \ln \Gamma(b) - \ln \Gamma(a+b)$$

95% Highest Density Interval (HDI)

discrete 200-step grid search accumulating the smallest interval containing 95% of posterior probability mass [θ_lower, θ_upper].

95%

Magnus-Tetens Dew Point Formulation

$$\alpha(T, RH) = \frac{17.27T}{237.7 + T} + \ln(RH/100)$$

$$T_dew = \frac{237.7\alpha}{17.27 - \alpha}$$

SYSTEM ARCHITECTURE

Telemetry and check-ins drive deterministic safety rules

[Open-Meteo GPS Sync] + [1-Tap Check-In] → [Adaptive Rules Engine] → [Dynamic Routine Card] → [Posterior Update]

1. High UV (≥ 6): strips photosensitizing daytime actives (AHAs, Retinoids) and mandates SPF 50+.
2. Arid Climate (<40% RH) / Tight Report: swaps gels for rich ceramide creams; prescribes PM “sandwich buffering”.
3. Humid Climate (>70% RH) / Greasy Report: swaps heavy creams for oil-free water-gels; prioritizes pore-clearing BHA/Niacinamide.
4. Stinging / Irritated Alert: triggers 72-hour emergency “Barrier Reset” (cleanser + soothing emollient only).

RESULTS

Two weather states produce two distinct routines



Test Case 1 — Humid Afternoon

33°C, 78% RH, UV 7.4 → foaming wash, 2% Niacinamide, oil-free gel, matte SPF 50+.



Test Case 2 — Arid Cold Evening

21°C, 31% RH → gentle wash, barrier rest, ceramide cream buffer.

Day 1: high epistemic uncertainty; 95% HDI is broad ([0.25, 0.72]).

Day 14: after logging 11 balanced days, posterior shifts right and tightens to [0.77, 0.89], with variance dropping by >65%.

NEXT STUDY LAYER

Future modules turn the prototype into a longitudinal study

Module 3 — Hypothesis Testing

Dynamic χ^2 test of independence comparing [Weather Regimes] × [Skin Complaints] to test H_0: Independence vs H_1: Dependence ($p < 0.05$).

Module 4 — Regression

Multiple Logistic Regression quantifying odds ratios (β coefficients) of ambient humidity and active strength on skin flare probability.

Module 5 — ANOVA

Two-Way Factorial ANOVA evaluating interaction effects between Routine Archetype (Factor A) and Seasonal Climate Regime (Factor B) on user satisfaction scores.

Data Persistence

SQLite/IndexedDB backend for longitudinal multi-month tracking.

CLOSING SUMMARY

DermaSync connects everyday utility to BAMAT207 rigor

1. Everyday Utility: replaces complex derm diagnostics with an effortless, 5-second adaptive daily companion.

2. Mathematical Rigor: employs exact closed-form Beta-Binomial conjugate updating with continuous log-space calculus.

3. Curriculum Alignment: covers BAMAT207 probability distributions, Bayes' Theorem, credible intervals, and planned hypothesis tests.

